Example 1. Let $S = \{\text{all cities in the U.S. except Washington D.C.}\}$, and

$$T = \{\text{all states in the U.S.}\}.$$

For each of the following
- Does the assignment define a function?
- If it does, what is the image?

(a) $f : S \to T$ assigns to each city the state it is in.

(b) $g : T \to S$ assigns to each state the largest city in it (by population).

(c) $h : T \to S$ assigns to each state the capital city of this state.

(d) $k : T \to S$ assigns to each state a city in that state with population over 5,000,000.

Example 2. (a) Which of the following define functions? Give a brief explanation - no formal proof required.
(b) If it is a map, is it surjective? Is it injective?

(a) $f : \mathbb{R} \to \mathbb{R}, \quad f(x) = \dfrac{x}{5}$

(b) $g : \mathbb{Z} \to \mathbb{Z}, \quad g(x) = \dfrac{x}{5}$

(c) $h : \mathbb{Q} \to \mathbb{R}, \quad h(x) = \dfrac{x}{5}$

(d) $\ell : \mathbb{R} \to \mathbb{Q}, \quad \ell(x) = \dfrac{x}{5}$

Example 3. Prove that the function $g : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$g(x, y) = (x + y, x - y).$$

is surjective.

Example 4. Prove or disprove: the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3 - x$ is injective.

Example 5. Fix $a, b \in \mathbb{R}$ with $a < b$ and consider the interval (a, b). Consider the map $f : (0, 1) \to (a, b)$ defined by $f(x) = a + (b - a)x$.

(a) Prove that f is surjective.

(b) Prove that f is injective.

Example 6. Let $f : \mathbb{Z} \to \mathbb{Z}$ be the mapping $f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ \frac{n+1}{2} & \text{if } n \text{ is odd.} \end{cases}$

Is f injective? Prove your answer.

Example 7. Consider the following map:

$$\begin{aligned} f : \mathbb{Z} \times \mathbb{Z} &\longrightarrow \mathbb{Z} \\ (m, n) &\longmapsto 3m - 4n \end{aligned}$$

(a) Prove that f is a function.

(b) Is f injective? Prove your answer.

(c) Is f surjective? Prove your answer.